Further Paleomagnetic Results from the Bloomsburg Formation[1]

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Paleomagnetic observations have been obtained from 161 specimens from the Bloomsburg redbeds (Upper Silurian) collected at 13 localities along the tectonic arc in the Valley and Ridge province in the Appalachians. The purpose was to determine if the curvature was original, or if it was produced by bending in plan, during, say, the Appalachian orogeny. Observations were obtained from the eastern and southern limbs and from the hinge area of the arc. Components of very high magnetic stability and substantial uniformity (mean D, I is 001°, −32°, precision $k = 35$, pole 32°N, 102°E) were found in 9 localities in the eastern limb and in the hinge area in Pennsylvania, indicating that the arc was original. In the southern limb, in Maryland and Virginia, however, less stable secondary components probably of Permian age appear to have replaced much of the original magnetization, so that a rigorous test was unfortunately not possible. Heating experiments showed that these secondary components could have been acquired during a period of moderate heating during burial.

Introduction. Thermal analysis of the Bloomsburg formation (Upper Silurian) sampled earlier [*Irving and Opdyke,* 1965] at Bloomsburg, Pennsylvania, showed that the fossil magnetization was comprised of two components, one (called the *thermally discrete* component) in which the blocking temperatures were confined to a temperature interval about 100°C below the Curie temperature (675°C) and the other, less stable component (called the *thermally distributed* component) in which the blocking temperature ranged from room temperature to about 500°C. The distributed component was directed along the Permian field direction, which was the field prevailing at the time that the beds were folded (Appalachian orogeny). It was generally found possible to remove the distributed component by thermal cleaning at 550°C, leaving the discrete component unaffected. The 'cleaned' directions had a similar inclination but a declination 30° greater than that observed by *Graham* [1949] in the Middle Silurian Rosehill formation, Maryland. This difference could be due to a change in the earth's field between the times of formation of the two outcrops or due to relative rotation of localities. Therefore, sampling along the arc in the eastern limb (localities A to D, M, and P, Figure 1), the hinge area (E, F), and the southern limb (G to L) was made in an attempt to decide between these possibilities. It would be pleasant to be able to report that the problem had been resolved in this way, but, in practice, difficulties arose that allowed only a partial answer to be obtained.

Sampling. Thirteen localities (A–M) were sampled (see the appendix and Figure 1), making 14 with the original Bloomsburg locality (P) [*Irving and Opdyke,* 1965]. The exposures consisted of fine-grained closely jointed red fine sandstones and siltstones. The sandstones could not be sampled; all samples were obtained from the siltstones. Several samples were taken from one or more beds at each locality. Each bed sampled is referred to as a site. Most samples were oriented drill cores. Drilling was sometimes impossible and oriented hand samples were

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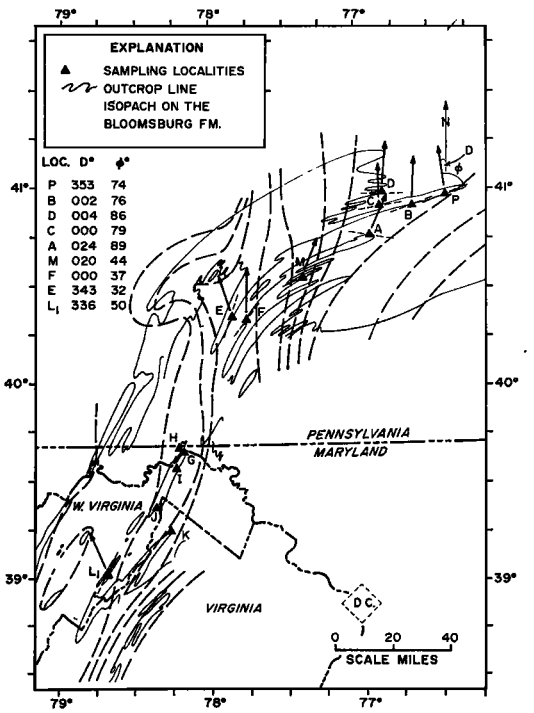


Fig. 1. Locality map with horizontal magnetic components $D°$. The dashed lines are the bedding strikes. ϕ is the angle between the strike angle and $D°$.

taken. All localities were in road cuttings. The samples were measured on the astatic magnetometer of *Roy* [1963*a*]. The directions were first referred to present horizontal and then were corrected for tilting by rotation about the strike. Magnetic stability of specimens from each locality and from most of the sites at a locality was studied by step-wise thermal demagnetization methods using an oven made by *Robertson* [1964]. A cleaning temperature of 550°C was used.

Results. All directions before and after thermal cleaning are plotted in Figure 2. The averages for sites and localities are listed in Table 1. There is a clear north and south grouping of directions. The *northerly group* (48% of the samples) shows substantial initial scatter but after cleaning is well grouped. The within-site scatter decreases after cleaning (Table 1). The thermal decay curves generally show a small increase up to 600°C (Figure 3, curve D) indicating the presence of two oppositely directed magnetizations of different stability. (See *Irving and Opdyke* [1965] for details.) The histograms of intensity values are skewed but become approximately symmetrical if logarithms are taken (Figure 4). The geometrical mean is considered more representative than the arithmetic mean. The intensities after cleaning (M_{550}) are higher than the initial ones (M_n) in the ratio $M_{550}/M_n = 1.1$. The *southerly group* comprises 32% of the samples. Cleaning does not greatly affect the scatter. The ratio M_{550}/M_n never exceeds unity and is usually about 0.6. Thermal cleaning does not produce any large displacement of the site mean directions. Finally, there is a less well defined *intermediate group* (20% of samples) whose directions after cleaning often have steep negative inclinations tending toward the northerly group. The ratio M_{550}/M_n averages 0.6 (Figure 4). Demagnetization usually causes a northerly migration of the directions; the displacements are sometimes small (Figure 5, site H2A) but are commonly large (Figure 6, site L2).

As in the previous work of *Irving and Opdyke* [1965], all the northerly directions are repeatable upon reheating in the range 550°–600°C. The southerly and intermediate group specimens are commonly repeatable up to this range but about a third are not; these specimens developed a very soft magnetization (IRM) apparently instantaneously if exposed to small dc fields (order 0.3 oe) and in a few extreme cases intensities of 10^{-5} emu were developed in this way and masked any useful remanence. In most specimens, however, these intensities did not exceed 2 to 3 $\times$ 10^{-7} emu. In a few southerly specimens measurable IRM effects were found at temperatures as low at 400°C.

Illustration of effects of thermal cleaning are given in Figure 5. Behavior is extremely variable. At site E1 there is little change. Site B5 illustrates the large northerly migration commonly encountered, and there is a large decrease in dispersion. A systematic but small migration occurring in intermediate directions is shown in site H2A. Site J1 illustrates the consistency generally found among southerly directions.

Distribution of groups geographically. Northerly directions predominate in the eastern limb and in the hinge area. Southerly directions predominate in the south limb. Usually the samples from one site show uniform properties.

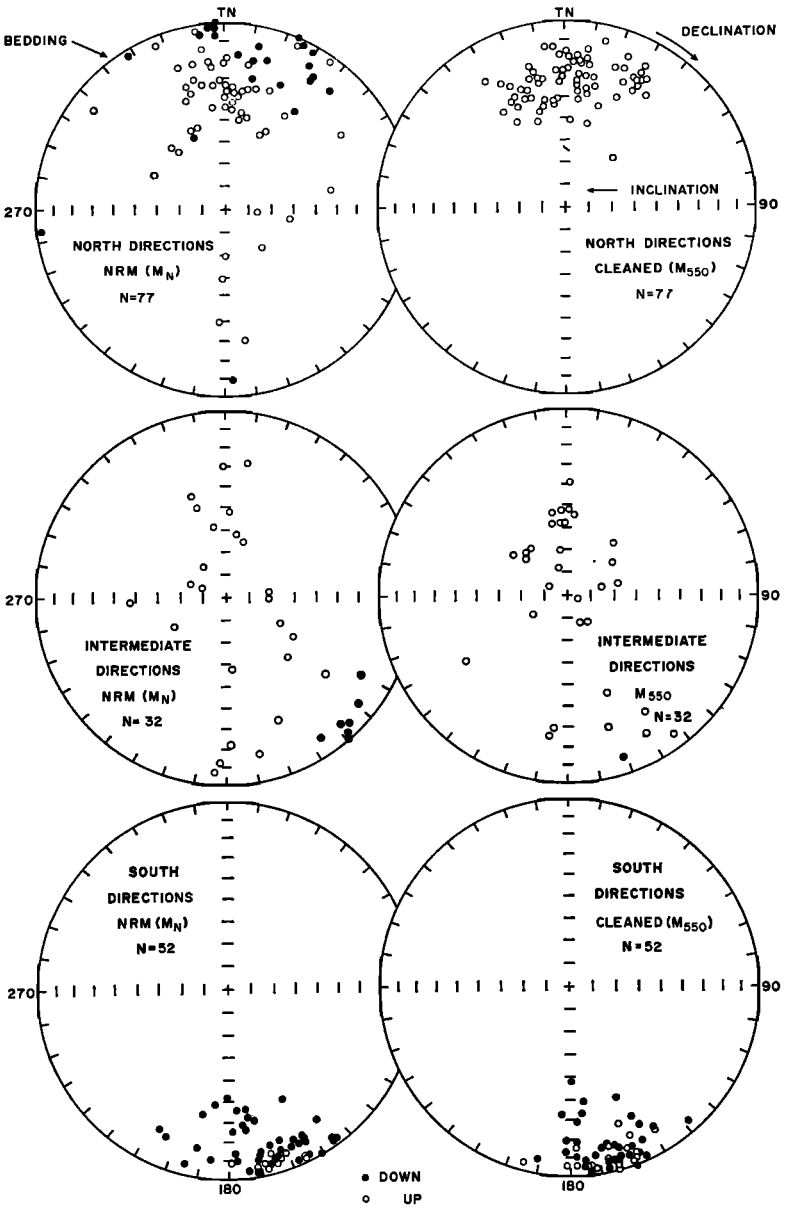


Fig. 2. Directions of magnetization before and after cleaning. The directions are plotted on equal area projections and are corrected for folding.

Sometimes, however, sites from the same locality belong to different groups, and sometimes there are intricate variations within a bedding level. For example, at locality K samples within a few centimeters of one another show comparable magnetic properties, but samples from different spots in the same bedding level over a small fold show either southerly (K1, K2, K4) or intermediate (K3) directions (Figure 7, left). At locality L (Figure 6) northerly (L1), intermediate (L2), and southerly (L3) are represented. At site H1 (Figure 8, left) specimens

TABLE 1. Mean Intensity and Directions at Each Site (Locality) before and after Thermal Cleaning at 550°C

Site Locality	N N_s	NRM M_N	NRM D_m, deg	NRM I_m, deg	NRM R	Cleaned M_{550}	Cleaned D_m, deg	Cleaned I_m, deg	Cleaned R	Pole	Group
A1	6	4.31	023	+27	5.49	4.0	022	−27	5.87	31°N, 077°E	N
A2	11	8.14	024	+11	10.69	8.29	027	−20	10.91	33°N, 071°E	N
A	**2**	**6.22**	**023**	**+19**	**1.87**	**6.30**	**024**	**−23**	**1.99**	**32°N, 074°E**	**N**
B2	4	3.83	312	−56	2.98[1]	6.37	357	−16	3.99	41°N, 107°E	N
B3	5	4.81	096	−63	3.61	5.84	355	−40	4.87	26°N, 108°E	N
B4	4	7.59	029	−45	3.94	9.19	022	−35	3.95	30°N, 102°E	N
B5	4	2.49	125	−77	3.35	3.82	010	−30	3.93	32°N, 092°E	N
B	**4**	**4.68**	**019**	**−72**	**3.12**	**6.30**	**001**	**−30**	**3.94**	**33°N, 102°E**	**N**
C1	4	5.43	001	−30	3.97	6.00	006	−24	3.98	37°N, 096°E	N
C2	2	6.67	352	−10	1.96	7.10	353	−33	1.99	31°N, 111°E	N
C	**2**	**6.05**	**356**	**−20**	**1.93**	**6.50**	**000**	**−28**	**1.98**	**34°N. 103°E**	**N**
D	**5**	**10.46**	**011**	**−39**	**4.94**	**11.90**	**004**	**−28**	**4.97**	**34°N, 098°E**	**N**
E1	11	6.72	353	−38	10.49	7.37	347	−42	10.92	25°N, 115°E	N
E2	4	6.97	347	−18	3.94	8.39	338	−26	3.96	33°N, 127°E	N
E	**2**	**6.34**	**350**	**−28**	**1.97**	**7.88**	**343**	**−34**	**1.98**	**29°N, 121°E**	**N**
F	**7**	**9.03**	**351**	**+00**	**6.87**	**5.12**	**000**	**−29**	**6.95**	**34°N, 102°E**	**N**
G_1	**4**	**4.66**	**176**	**+34**	**3.97**	**2.73**	**172**	**+30**	**3.93**		**S**
G2	5	5.36	172	−27	4.75	2.89	191	−54	4.19		I
G3	7	8.06	088	−72	6.57	5.51	025	−68	6.71		I
H1A	5	5.71	264	−69	3.38	4.55	322	−72	4.93		I
H1B	5	5.36	196	+14	4.94	2.89	179	+03	4.82		S
H2A	2	10.73	292	−76	2.00	6.90	315	−65	2.00		I
H2B	5	7.16	157	+02	4.91	3.80	153	+04	4.96		S
$H_{(1,2)B}$	**2**	**6.26**	**176**	**+08**	**1.88**	**3.35**	**166**	**−04**	**1.95**		**S**
I	**5**	**4.63**	**166**	**−03**	**4.95**	**2.40**	**172**	**−07**	**4.91**		**S**
J1	6	8.85	160	+08	5.88	5.35	163	−03	5.77		S
J2	4	8.58	165	+08	3.96	6.85	163	+00	3.94		S
J3	6	6.46	169	+02	5.90	3.56	169	−01	5.94		S
J	**3**	**7.96**	**164**	**+06**	**2.99**	**5.25**	**165**	**+01**	**2.99**		**S**
K1	4	7.17	171	+24	3.93	4.70	175	+10	3.99		S
K2	3	7.76	166	+30	2.97	5.18	179	+33	2.93		S
K4	2	6.31	189	+36	2.00	4.79	181	+35	2.00		S
$K_{(1,2,4)}$	**3**	**7.08**	**175**	**+30**	**2.96**	**4.89**	**178**	**+26**	**2.94**		**S**
K3	5	31.44	351	−46	4.90	25.80	356	−50	4.96		I
L_1	**7**	**4.84**	**350**	**−36**	**6.84**	**5.39**	**336**	**−37**	**6.97**	**26°N, 126°E**	**N**
L2	7	3.07	134	+03	7.68	1.56	156	−33	6.83		I
L_3	**8**	**11.21**	**156**	**+06**	**7.87**	**6.68**	**159**	**+04**	**7.81**		**S**
M	**4**	**1.97**	**189**	**−44**	**2.04**	**1.55**	**020**	**−35**	**3.76**	**27°N, 082°E**	**N**
P1	7		183	−66	6.69		359	−42	6.82	25°N, 106°E	N
P2	3		353	−35	2.99		352	−24	3.00	36°N, 115°E	N
P3	8		358	−46	7.69		000	−35	7.96	30°N, 106°E	N
P4	3		342	−41	2.95		344	−32	2.98	30°N, 123°E	N
P5	6		150	−42	5.78		350	−36	5.85	28°N, 116°E	N
P6	7		064	−45	4.73		357	−29	6.94	33°N, 109°E	N
P7	6		333	−40	5.69		339	−22	5.96	34°N, 131°E	N
P8	7		000	−39	6.95		355	−34	6.97	30°N, 111°E	N
P10	4		355	−26	3.98		357	−26	4.00	36°N, 109°E	N
P11	1		148	−63	...		348	−36	...	28°N, 118°E	N
P12	5		020	−39	4.87		355	−26	4.99	35°N, 111°E	N
P13	2		352	−44	2.00		358	−29	2.00	33°N, 108°E	N
P14	4		090	−12	2.53		351	−36	3.82	29°N, 116°E	N
P15	4		351	−23	3.98		347	−24	3.96	35°N, 121°E	N
P16	1		122	−16	...		022	−40	...	23°N, 084°E	N

TABLE 1. (Continued)

Site Locality	N N_s	NRM M_N	NRM D_m, deg	NRM I_m, deg	NRM R	Cleaned M_{550}	Cleaned D_m, deg	Cleaned I_m, deg	Cleaned R	Pole	Group
P17	4		351	−42	3.96		347	−23	3.98	35°N, 121°E	N
$P_{(1-17)}$	**16**		**020**	**−56**	**11.63**		**353**	**−31**	**15.76**	**32°N, 113°E**	**N**
P9	2		177	+05	1.99		186	−09	1.76		S
P18	2		181	−26	1.99		331	−61	1.99		I

Notes:
The NRM of one specimen was not measured.
Bold faced characters indicate a locality mean direction. Classification into groups is given in the right-hand column. A subscript means that the locality has mixed site mean directions and indicates the sites that belong to either the northerly or the southerly group. An absence of subscript means that all site mean directions at that locality belong to the same group.
N = number of specimens; N_s = number of sites.
M_n and M_{550} are the NRM intensity and the remaining intensity after heating to 550°C; the arithmetic means are given in 10^{-6} emu.
R is the magnitude of the resultant vector.
Poles are given for cleaned northerly directions.

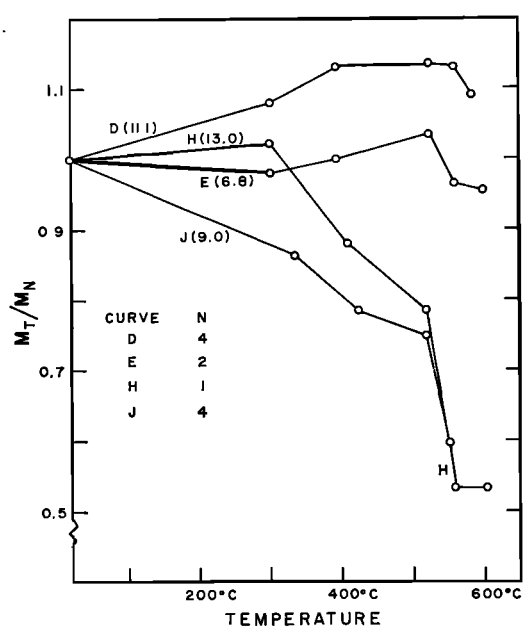


Fig. 3. Thermal decay of intensity normalized to unity at room temperature. The NRM intensity is given in parentheses in emu/cc × 10^{-6}. N is the number of specimens contributing to each curve. D and E belong to the northerly group, H to the intermediate group, and J to the southerly group.

from a few centimeters apart in the same level show southerly (H1B) and intermediate (H1A) directions.

Response to fold test. Locality mean directions (after cleaning) are plotted in Figure 9. With respect to horizontal the northerly directed localities are widely scattered ($N = 9$, $R = 7.3$, $k = 5$). When corrected for dip, the precision is much improved ($R = 8.8$, $k = 35$). Therefore, the northerly directed magnetization predates the folding.

The over-all grouping of the southerly directed localities shows an improvement after correction ($N = 6$, $R = 5.9$, $k = 25$) compared with that in situ ($R = 5.6$, $k = 14$), although the improvement is less marked than for the northerly directions. There are often large variations of bedding within localities. At locality K, sites K1, K2, and K4 show a marked improvement after correction (Figure 7), indicating a prefolding magnetization. A similar effect is also shown by some groups of specimens (H1B and H2B) at locality H (Figure 8, right). Therefore, it appears that most of the southerly directed magnetization, like the northerly one, was acquired before folding, although the result is much less clear cut.

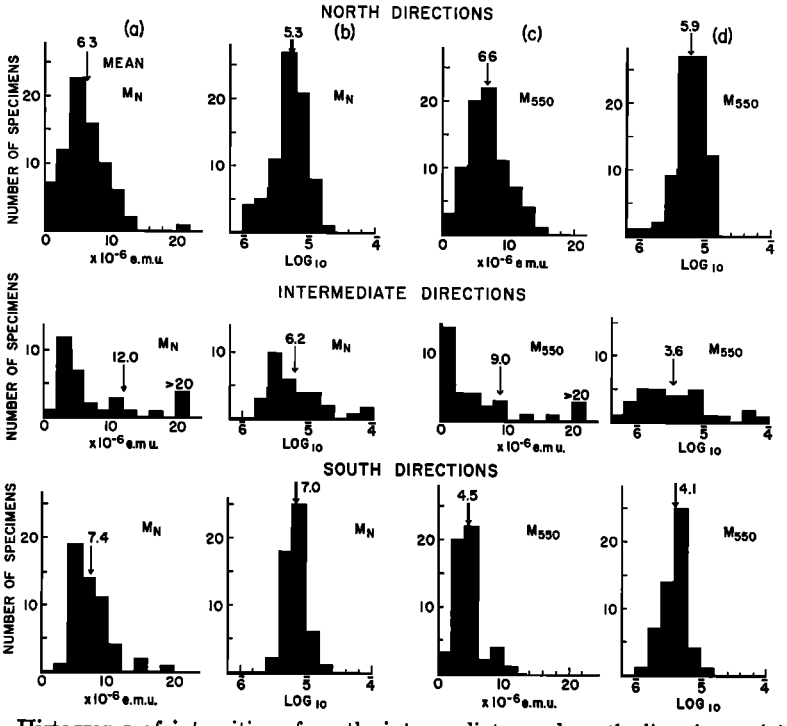


Fig. 4. Histograms of intensities of north, intermediate, and south directions: (*a*) NRM, (*b*) $\log_{10}$ NRM, (*c*) after heating to 550°C, and (*d*) $\log_{10}$ after heating to 550°C. The arithmetic and geometrical means are indicated.

Heating experiments. Most of the cleaned southerly directions are like those in Permian rocks of West Virginia [*Helsley*, 1965] and eastern Canada [*Roy*, 1963*b*, 1966; *Black*, 1964]. Since this was the time in which the Bloomsburg beds were deformed (Appalachian orogeny), it is reasonable to suppose that the southerly magnetizations were acquired at that time, either by chemical change, by viscosity during a period of moderate heating at low temperatures (viscous PTRM as referred to by *Chamalaun* [1964]), or a combination of the two. Heating tests comparable to those made by *Shimizu* [1960] and *Briden* [1965] were made to investigate viscous PTRM effects. The procedure was to heat specimens at 300°C in the earth's field for periods of up to 3 weeks and then to measure the magnetization acquired in that time. To simplify the experiments only vertical components were measured. The vertical component of the applied field was 0.56 oe. This may not be a realistic modeling of conditions during burial, but the experiments give some indication of the effects likely to arise.

An automatically controlled nonmagnetic furnace built at the Dominion Observatory was used. Its description is given here for the first time. The main heater consists of a ceramic tube (length 61.3 cm, inner diameter 12.7 cm, outer diameter 14.0 cm) carrying on its outer surface a primary nichrome (swg 11, length 2.9 meters, resistance 7.5 ohms) winding, which is looped lengthwise to be noninductive. A secondary trimming heater (swg 30, resistance 100 ohms) is located just inside the ceramic tube. Insulation is provided by sectorial ceramic bricks and circular bungs at the ends. The over-all diameter is 40 cm. The primary winding will carry up to 18 amp and temperatures of 1100°C can be attained. The primary winding is set to constant current and provides 90% of the heat required for a given temperature. The remainder is provided by the secondary heater controlled automatically by a proportional temperature controller that can be preset to the desired temperature. The secondary heater has a rapid response and prevents temperature fluctuation, which would normally occur in such

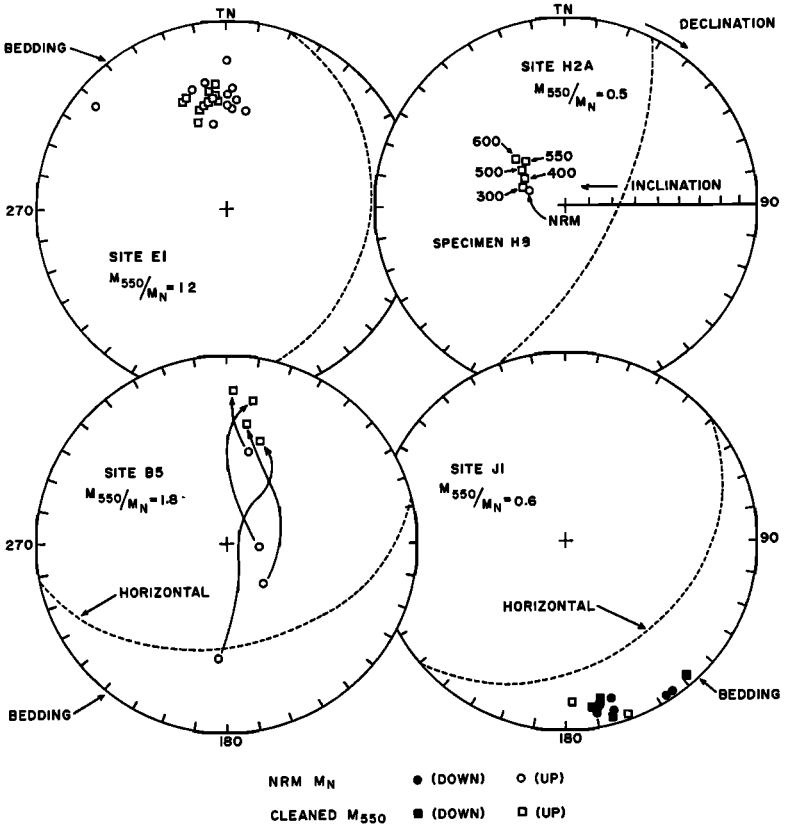


Fig. 5. Examples of thermal cleaning plotted on an equal area polar net.

large capacity furnaces. The temperature is maintained to ±0.5°C. Temperatures are measured by a platinum resistance sensor and a platinum −10% rhodium thermocouple.

At first, it was thought necessary to work with complete demagnetized specimens, since any magnetization initially present might be affected by heating and make it impossible to measure viscous buildup. In the first experiment, therefore, powder specimens were employed. It was found, however, that, although duplicate powder samples (>325 mesh) from the same sieve fraction gave results repeatable to within 3%, the results from powders of smaller grain size (<325 mesh) gave intensities 2 or 3 times greater. The specific intensities (range 30 to 150 $\times$ 10^{-6} emu) of powder, when heated at 300°C for even only 10 minutes, were very much greater than the observed NRM. The effect is clearly undesirable, and the powder samples were abandoned and natural rock used instead.

Specimens previously demagnetized at 500°C (M_{550}) were first heated for 3.8 days at 300°C in the earth's field, then measured (circle 1 in Figures 10, 11, and 12), and heated for 14.8 days and remeasured (circle 2). Values of the magnetization produced during heating were then obtained by subtracting the vertical component of M_{550} (present before prolonged heating) from that measured afterward. The specimens were then thermally demagnetized again at 550°C, and a new run started, heating first for 6 minutes, 16 minutes (crosses 3 and 4 in Figures 10, 11, and 12), and so on up to 10 days (cross 9). Finally the specimens were demagnetized for a third time at 500°C. This procedure is justified by the fact that after each prolonged heating the original magnetization M_{550} was retrieved in all 10 specimens studied; that is, the M_{550} remained unchanged throughout the thermal cycling. For example, the directions (D, I) and intensities M_{550} of

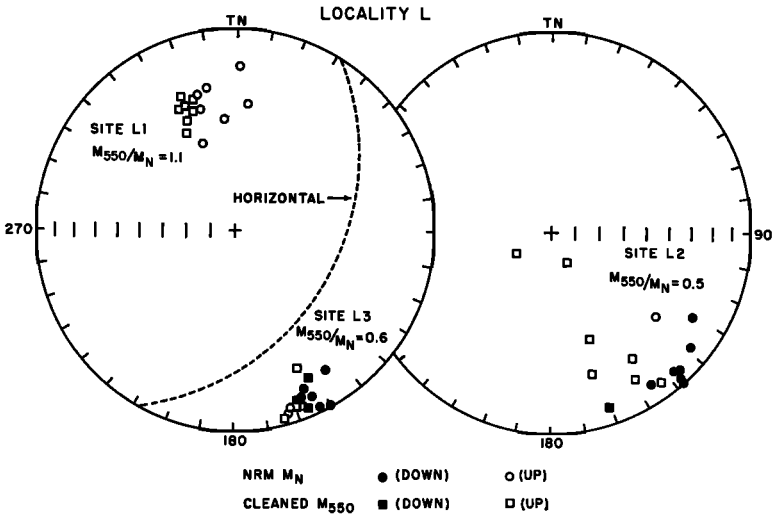


Fig. 6. Effect of thermal demagnetization on three sites from the same locality, representing each group: northerly (L1); intermediate (L2); southerly (L3). Equal area polar net.

specimen D2 before the first buildup run, between the first and second run, and after the second and final run are, respectively, 181°, −14°, 12.0 × 10^{-6} emu; 180°, −16°, 11.5 × 10^{-6} emu; 180°, −16°, 11.5 × 10^{-6} emu. Clearly the original fear that buildup might not be observable because of possible thermally induced changes affecting pre-existing components was unfounded.

The buildup is roughly proportional to the logarithm of time, although there is a tendency to level off in some specimens, indicating an approximation to a limiting saturation value. The intensities produced are comparable in magnitude to the NRM. In most specimens studied (Figures 10 and 11) the results are repeatable within close limits (compare circles 1 and 2 with crosses 8 and 9), and negligible changes of the viscous component are induced by heating. In three specimens, however, Fig-

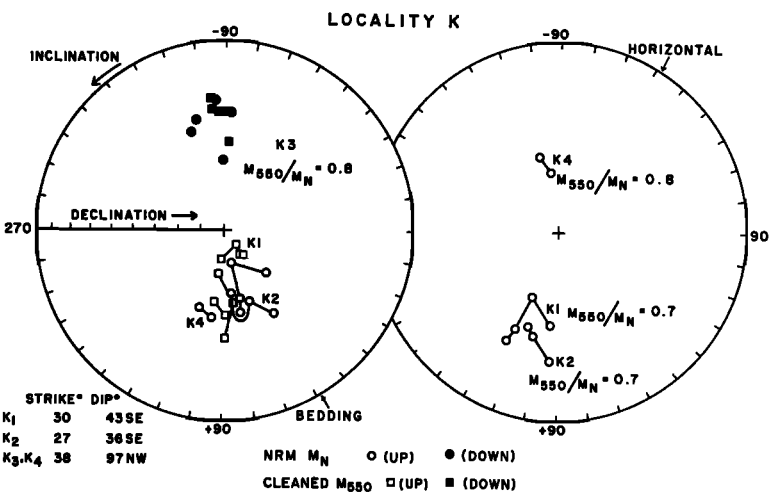


Fig. 7. Equal area equatorial net. Up (down) indicates directions from 90° to 270° (270°–0°–90°). Directions plotted w.r.t. bedding (left) and horizontal (right). K1, K2, K4 are southerly directions and show positive response to tilt test. K3 is an intermediate direction.

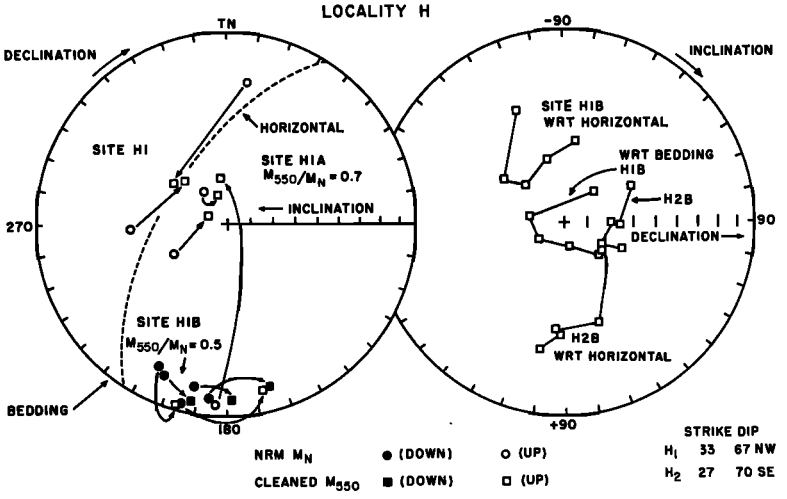


Fig. 8. On the left (equal area polar projection) directions at site H1 showing separation into southerly (H1B) and intermediate (H1A) directions. On the right (equal area equatorial projection) the effect of correcting for folding on the southerly directions (after cleaning) of H1B and H2B.

ure 12), the results were not repeatable; the second buildup gave values 25% higher than the initial run. Throughout these severe heating experiments, however, the component M_{550} of the fossil remanence was unchanged and it is clearly of very great stability.

Summary and discussion. The northerly directed magnetization consists of two readily separated components. There is a hard thermally discrete component acquired before folding. Its mean and corresponding pole are given in Table 2. This hard component is overlain by a softer distributed magnetization that is directed approximately along the Permian field and that can be removed by thermal cleaning. An artificial component of comparable magnitude can be built up in these specimens by heating at 300°C for 3 weeks. It is therefore

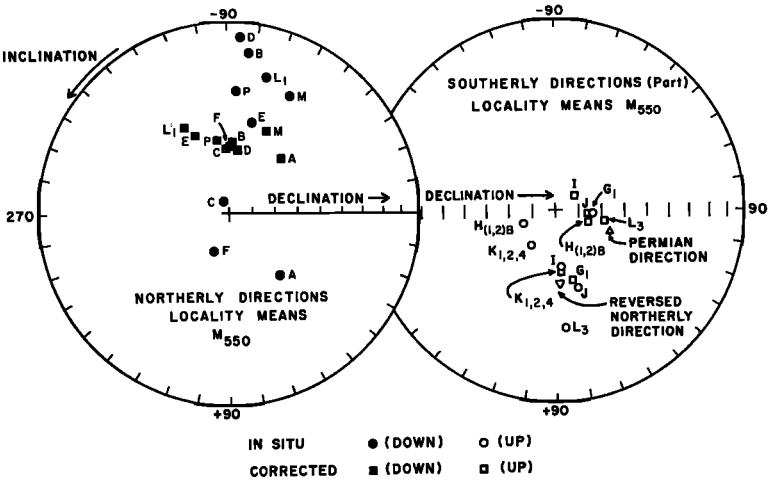


Fig. 9. Effect of bedding on locality mean directions. The subscript, which means that the locality has mixed site mean directions, indicates the sites that belong to the northerly or southerly group. An absence of subscript means that all site mean directions at that locality belong to the same group. Up (down) indicates directions from 90° to 270° (270°–0°–90°).

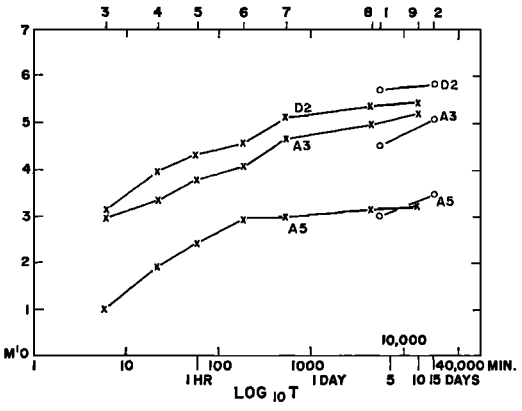


Fig. 10. Viscous PTRM. Repeatable specimens of the northerly group. Order in which results were obtained is indicated above. The circles denote results from the first run; the crosses, results from the second run. Intensities (M_1) given in 10^{-6} emu.

concluded that the distributed component is Permian and was acquired at a time of deep burial during the Appalachian orogeny, when the earth's field appears to have maintained a positive (reversed) polarity for a very long time.

The southerly directed magnetization is mostly thermally distributed. Because the directions change little during demagnetization, it is impossible to say whether it consists of a single distributed component or of a large thermally distributed component superimposed

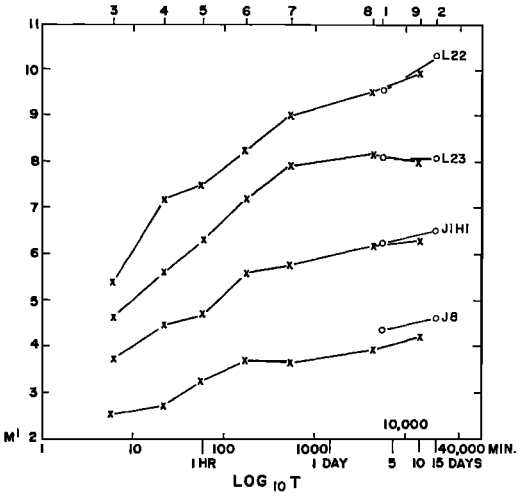


Fig. 11. Viscous PTRM. Repeatable specimens of the southerly group. Legend as for Figure 10.

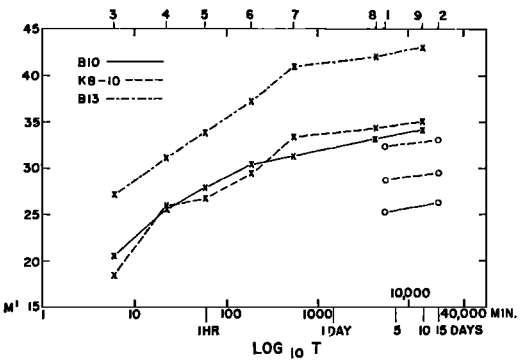


Fig. 12. Viscous PTRM. Nonrepeatable specimens from northerly (B10, 13) and southerly (K8) group. Legend as for Figure 10.

on a small thermally discrete one. The southerly directions fall roughly into two groups: localities K(1, 2, 4) and G1 (mean 175°, +28°), which are anti-parallel to the northerly group, and localities I, J, H(1, 2B) and L3 (mean 165°, +01°), which are close to the Permian field. The former group can therefore be reversed 'Silurian' directions; the latter, Permian. Heating experiments show that much of the distributed component of these southerly directed magnetization could be due to viscous PTRM acquired during deep burial; yet, they do retain a substantial southerly component even after cleaning. These southerly directions are found mostly in the southern limb of the arc (Figure 1) in which deformation is greater than in the northern limb. It is therefore possible that the degree and duration of heating has been such as to make over the entire magnetization into a Permian direction, whereas, in the less deformed northern limb, the process has only proceded part of the way, leaving unaffected the underlying discrete component. Clearly the results obtained so far provide little of the detail of these processes, but it seems fairly certain now that the bulk of these southerly directed magnetizations, both those comprising part of the fossil remanence of the northern limb specimens and those making up the entire remanence in most of the southern limb specimens, were acquired during the late Paleozoic.

The intermediate group tends upon demagnetization to move toward, but never quite reach, the northerly group. This is interpreted as being due to incomplete cleaning.

The thermally discrete component from the

TABLE 2. Summary of Northerly Group Directions

Area	Locality	N	D_m, deg	I_m, deg	R	k	Pole Position
North limb	A-B-C-D-P	5	004	−28	4.93	58	34°N, 097°E
Hinge	E-F-M	3	001	−34	2.93	27	31°N, 102°E
South limb	L_1	1	336	−37	1.00	—	25°N, 127°E
Total		9	001	−32 (8.8°)*	8.77	35	32°N, 102°E (6°, 10°)†

* Fisher's circle of confidence ($P = 0.05$).
† Model A polar errors δ_p and δ_m ($P = 0.05$).

eastern limb and the hinge is very stable and may be used for a tentative structural interpretation. The directions after correction for dip (squares in Figure 9) have inclinations agreeing better than declinations. Within this distribution the sites have no discernible order, which means that it cannot be explained by simple bending of the Appalachian arc. It could result from second-order imperfections in the correction procedure (simple rotation about the strike may be an incomplete restoration). The horizontal components (after correcting for folding) are given in Figure 1 together with the analysis of bedding strikes and declinations. It is clear that the angle (ϕ) between the horizontal component and strike does not indicate that there was any bending of the arc; otherwise ϕ would be approximately constant. In particular, localities E and F at the hinge of the arc suggest that no bending has occurred, ϕ being 30° or 40° less than expected. These observations do not support the idea of bending in plan. The declination of 35° at site L1 seems to indicate some anticlockwise rotation, but no statistical significance can as yet be ascribed to it.

If, as is just possible, the directions at K(1, 2, 4) and L3 are reversed 'Silurian,' this would indicate strongly that the curvature was original. The separation of these directions from secondary Permian ones cannot, however, be made confidently. The main hope for a satisfactory tectonic test now seems to rest on the application of better cleaning techniques to the intermediate group. With improved apparatus now under construction it may be possible to retrieve the northerly component that appears to be present in the intermediate group.

Although the results are less decisive tectonically than had been expected, they do define a component of high stability and substantial uniformity for the 9 Pennsylvanian localities (Table 2).

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Appendix

Reference list of collecting localities; all bearings are east of true north:

A. Outcrop in a barrow pit along a farm lane, 5 km west of Selinsgrove, Pennsylvania, 40°48′N, 076°55½′W; samples from two levels 1.5 meters apart, A1 (6 cores) dipping 38° at 19° and A2 (3 hand samples, 11 cores) dipping 40° at 16°.

B. U. S. route 11, 5 km west of Danville, Pennsylvania, 40°55′N, 076°40′W; samples from five levels spanning 140 meters; B1 (3 cores) and B2 (4 cores) dipping 46° at 166.5° and B3 (5 cores), B4 (4 cores), and B5 (4 cores) dipping 45° at 167.5°.

C. Along the west bank of the Susquehanna River, Lewisburg, Pennsylvania, 1 km south of Buknell University 40°57′N, 076°52½′W; samples from two levels 1 meter apart and dipping 23° at 349°; C1 (4 cores) and C2 (2 cores).

D. Highway, U. S. route 15, 0.5 km north of West Milton, Pennsylvania, 41°02′N, 076°52′W; samples (5 cores) from one level dipping 54° at 180°.

E. Highway U.S. route 22 on the north side of the Juniata River across from the town of Mount Union, Pennsylvania, 40°23½′N, 077°53′W; samples from two levels 1.5 meters apart; E1 (11 cores) dipping 26° at 107° and

E2 (2 hand samples, 4 cores) dipping 28° at 103°.

F. Highway 103, 25 km south of Licking Creek and 5 km south of Newton Hamilton bridge, Pennsylvania, 40°21′N, 77°50′W; four samples (10 cores) from one level dipping 66° at 307.

G. Railway cut along Western Maryland R.R. 3.2 km west of Hancock, Maryland, 39°40′N, 078°14′W; samples from three sites in the same level; G1 (4 cores) dipping 40° at 303°, G2 (4 cores and 1 hand sample (1 core)) dipping 55° at 135°, and G3 (9 cores) dipping 63° at 136°.

H. Highway 40 west of Hancock, Maryland, 3.7 km from bridge across Potomac 39°42′N, 078°13′W; samples from two sites in same level; H1 (10 cores) dipping 67° at 303° and H2 (7 cores) dipping 70° at 117°.

I. Highway 9 between Great Cacapon and Berkely Springs 39°38′N, 078°14′W; 5 cores from one level dipping 49° at 123°.

J. Outcrop in the village of Bloomery, West Virginia, 39°22′N, 078°22′W; samples from three levels spanning 10 meters; J1 (4 cores and 1 hand sample (2 cores)), J2 (4 cores), and J3 (4 cores and 1 hand sample (2 cores)) all dipping 36° at 141°.

K. Highway 50, 5 km east of Gore, Frederick County, Virginia, 39°16′N, 078°18′W; samples from four levels spanning 17 meters; K1 (4 cores) dipping 43° at 120°, K2 (4 cores) dipping 36° at 117°, and K3 (7 cores) and K4 (2 hand samples (3 cores)) both dipping 97° at 308°.

L. Highway 55 at Lost River east of McCawley, Hardy County, West Virginia, 39°01′N, 078°43′W; samples from three levels spanning 160 meters; L1 (6 cores and 1 hand sample (1 core)) dipping 49° at 116° and L2 (6 cores and 1 hand sample (2 cores)) and L3 (8 cores and 1 hand sample (1 core)) both dipping 48° at 122.5°.

M. U.S. route 22, 130 meters north of the Mifflintown (Pennsylvania) town line; 40°34.5′N, 077°24′W; 6 cores from one level dipping 24° at 154.5°.

P. Fishing Creek, Bloomsburg, Pennsylvania [*Irving and Opdyke*, 1965].

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